UNIVERSITY AT ALBANY
State University of New York

College of Arts and Sciences
Department of Mathematics and Statistics

Applied Topology in Albany (ATiA) Seminar

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Persistent fundamental groups and lower-central-quotient barcodes

Friday, November 13, 2026
12:00 p.m. Hudson 0110.
Zoom ID: 812 1947 8289 - Passcode: 233444

Abstract. We present an efficient algorithm for computing the persistent fundamental group of a filtered CW-complex in terms of group presentations. From these presentations, we derive persistent invariants associated with the lower central series and obtain a family of barcodes that capture topological features not detected by persistent homology. This framework extends the scope of topological data analysis by enabling the study of non-abelian aspects of the fundamental group from filtered data.